

Arrays in Javascript

In javascript size of array grows or shrinks dynamically.

Array allows to store heterogeneous data

To create a array

```
var a=new Array(2,3,4,"xxxxx")
```

```
var b=[1,2,3,"xxxxx"]
```

to create a copy of the array

```
var c=[1,2,3,...a] //... is a spread operator of array, length of c is 7
```

To copy the reference

```
var d=[1,2,3,a] //length of d is 4
```

```
a=[1,2,3,4]
```

```
b=[10,20,a]
```

```
c=[11,22,...a]→[11,22,1,2,3,4]
```

arr.push(val)	to add the value at the end of the array	
arr.pop()	to delete the value from the end	
arr.unshift(val)	to add the value at the beginning	
arr.shift()	to delete the value from the beginning	
arr.splice(position, number)	delete the number of elements starting from the given position	arr=[1,2,3,4,5,6] arr.splice(3,2) //4 and 5 will be deleted [1,2,3,6]
arr.splice(position, number,list of values)	delete the number of elements starting from the given position, and replace it with the list of values	arr=[1,2,3,4,5,6] arr.splice(3,2,100,200,300,400,500) //4 and 5 will be replaced by the list of values [1,2,3, 100,200,300,400,500,6]

arr.splice(position, 0,list of values)	to add the values at the given position	arr=[1,2,3,4,5,6] arr.splice(3,0,100,200,300,400,500) //all the values will be added at the position [1,2,3, 100,200,300,400,500,4,5,6]
arr.indexOf(value)	find the position of the first occurrence of the given value This function is useful when the value is known	arr=[12,13,12,15,13] arr.indexOf(13) 1 arr.indexOf(130) -1
arr.findIndex(predicate function)	predicate function -> accepts one parameter and returns true / false findIndex function will find the position of the value for which predicate function returns true	arr=[12,13,12,15,13,20] //to find index of first value which is divisible by 5 //findindex will give the index of 15, arr.findIndex((val,index,arr)=>val%5==0) //use return keyword inside {} arr.findIndex((val,index,arr)=>{return val%5==0}) arr.findIndex(val=>val%5==0)
arr.find (predicate function)	predicate function -> accepts one parameter and returns true / false find function will find the value in the array for which predicate function returns true	arr=[12,13,12,15,13,20] //to find the first value which is divisible by 5 //find will give the value 15, arr.find ((val,index,arr)=>val%5==0) //use return keyword inside {} arr.find ((val,index,arr)=>{return val%5==0}) arr.find (val=>val%5==0)
arr.filter(predicate function)	predicate function -> accepts one parameter and returns true / false filter function will find the all	arr=[12,13,12,15,13,20] //to find all values which is divisible by 5 //filter will give the array of all values which are divisible by 5 arr.filter((val,index,arr)=>val%5==0) //use return keyword inside {}

	the values for which predicate function returns true	<pre>arr.filter((val,index,arr)=>{return val%5==0})</pre> <pre>arr.filter(val=>val%5==0)</pre>
arr.map(coverter function)	map function will apply the given expression on every value in the array and return a new value filter function will find the all the values for which predicate function returns true	<pre>arr=[12,13,12,15,13,20]</pre> <pre>//to find squares of all the numbers</pre> <pre>arr.map((val,index,arr)=>val*val)</pre> <pre>//use return keyword inside {}</pre> <pre>arr.map((val,index,arr)=>{return val*val})</pre>
Arr.reduce(function,[initial value])	To convert many values into one value use reduce Ex: to find sum of all numbers To find a string with maximum length	<pre>strarr=["Python","perl","linux","OS","Javascript"]</pre> <pre>// to concatenate first 3 letter of each string</pre> <pre>ans=strarr.reduce((acc,s)=>acc+s.slice(0,3),"")</pre>
Arr.sort([function])	Sort function always sorts data based on ascii values Hence to do numeric sorting we need to pass a function	<pre>function compare(a,b){</pre> <pre> //if(a<b)</pre> <pre> // return -1;</pre> <pre> //else if(a===b)</pre> <pre> // return 0</pre> <pre> //else</pre> <pre> // return 1;</pre> <pre> return a-b;</pre> <pre>}</pre> <pre>document.write(arr.sort(compare)+"
")</pre> <pre>document.write(arr.sort((a,b)=>a-b)+"
")</pre>

`arr=[10,20,30]`
`arr1=[1,2,3]`
`arr=[10,20,30, arr1]` `[10,20,30,]`

`arr1=[10,20,30, ...arr1]` `10,20,30, 1,2,3`

`var arr=[2,3,12,4,5,6]`
`arr.filter(val=>val%2==0)`
`2, 12, 4, 6`

`var arr=[2,3,12,4,5,6]`
`arr.map(val=>val*val)`
`4, 9, 144, 16, 25, 36`

`var arr=[2,3,12,4,5,6]`
`arr.reduce((acc, val)=>acc+val)`

acc	val
2	3
5	12
17	4
21	5
26	6
32	

`var arr=[2,3,12,4,5,6]`
`arr.reduce((acc, val)=>acc+val, 100)`

acc	val	acc	val
100	2	117	4
102	3	121	5
105	12	126	6
117		132	

Storage API

Now a days instead of cookies we use `localStorage` or `sessionStorage`, these objects are browser specific

The properties stored in `localStorage` will remain, even after you close browser window, till you delete it

The properties stored in `sessionStorage` will remain, till you close browser window,

The data in `localStorage` gets stored in string format

Read data from <code>localStorage</code>	<code>localStorage.count</code>	<code>localStorage.getItem("count")</code>
To assign value to a property	<code>localStorage.count=10</code>	<code>localStorage.setItem("count",10)</code>
To delete one of the properties		<code>localStorage.removeItem("count")</code>
To delete all properties		<code>localStorage.clear()</code>

geolocationAPI

It helps to find your current location or live location